MIE286 Lecture Notes After Midterm

Probability and Statistics

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Chapter 1

Chapter 8: Fundamental Sampling Distributions and Data Descriptions

1.1 Key Definitions

Population: The totality of observations with which we are concerned. In other words, it is the entire group of individuals or measurements of interest. We often cannot observe the entire population, so we rely on samples to make inferences.

Sample: A subset of a population. We collect data on the sample and use it to draw conclusions about the population.

Parameter: A number that describes the population. Parameters are fixed but in practice unknown. For example, the population mean μ and variance σ^2 are parameters. Since we rarely observe the whole population, we estimate parameters using statistics.

Statistic: A function of sample data. It is itself a random variable with a distribution. For example, the sample mean $\bar{X}$ and sample variance S^2 are statistics. Because a statistic depends on which sample happens to be drawn, different samples produce different values of the statistic.

The key insight is that a statistic is a random variable. This means we can talk about its probability distribution, which leads to the concept of a sampling distribution.

1.2 Random Samples

Definition: Simple Random Sample

The random variables $X_1, X_2, \ldots, X_n$ are said to form a **(simple) random sample** of size n if the X_i's are **i.i.d.** (independent, identically distributed), with each X_i having probability distribution $f(x)$.

Because the X_i are independent and identically distributed, the joint distribution factors as:

$$f(x_1, x_2, \ldots, x_n) = f(x_1)\, f(x_2) \cdots f(x_n).$$

Why independence matters: Independence means that the value of one observation does not affect the probability of any other observation. This is a reasonable assumption when the population is very large relative to the sample, or when sampling is done with replacement. In practice, even sampling without replacement from a large population is approximately independent.

The sample mean is denoted $\bar{X}$ and computed as:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Conceptual picture: Imagine taking many samples of size n from the population. From each sample, compute $\bar{X}$. Each sample gives a different $\bar{X}$. The collection of all these sample means forms a distribution.

X_1	X_{11}	X_{21}	$\cdots$	$X_1 \sim f(x)$
$\vdots$	$\vdots$	$\vdots$		
X_n	X_{1n}	X_{2n}		$X_n \sim f(x)$
$\bar{X}$	$\bar{X}_1$	$\bar{X}_2$		$\bar{X}$ is a r.v. $\longrightarrow$ Sampling Distribution

Each column represents a separate sample drawn from the same population. The bottom row shows the sample mean computed for each sample. The distribution of all these sample means is the *sampling distribution* of $\bar{X}$.

1.3 Sampling Distribution

Definition: Sampling Distribution

The probability distribution of a statistic is called a **sampling distribution**. It describes the variability of the statistic across all possible samples of the same size from the same population.

The sampling distribution depends on:

- the size of the population,
- the size of the sample, and
- the method of sampling.

Why sampling distributions matter: The sampling distribution tells us how much a statistic (like $\bar{X}$) would vary from sample to sample. This is the foundation for making inferences about population parameters—it allows us to quantify uncertainty.

1.4 Sampling from a Normal Distribution

1.4.1 Theorem: Distribution of the Sample Mean from a Normal Population

Theorem (Normal Population)

Suppose a random sample of size n is taken from $N(\mu, \sigma^2)$. That is,

$$X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu,\, \sigma^2).$$

Each observation X_i, $i = 1, 2, \ldots, n$, has the same distribution. Then:

$$\bar{X} \sim N\left(\mu,\ \frac{\sigma^2}{n}\right)$$

and equivalently,

$$\sum_{i=1}^{n} X_i \sim N(n\mu,\ n\sigma^2).$$

This was shown through moment generating functions (mgf's). Recall that the mgf of $N(\mu, \sigma^2)$ is $M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)$.

Proof (via mgf's):

The mgf of a single $X_i \sim N(\mu, \sigma^2)$ is:

$$M_{X_i}(t) = \exp\left(\mu t + \tfrac{1}{2}\sigma^2 t^2\right).$$

For the sum $\sum_{i=1}^{n} X_i$, since the X_i are independent:

$$M_{\sum X_i}(t) = \prod_{i=1}^{n} M_{X_i}(t) = \left[\exp\left(\mu t + \tfrac{1}{2}\sigma^2 t^2\right)\right]^n = \exp\left(n\mu t + \tfrac{1}{2}\, n\sigma^2 t^2\right).$$

This is the mgf of $N(n\mu,\ n\sigma^2)$, so $\sum_{i=1}^{n} X_i \sim N(n\mu,\ n\sigma^2)$.

For the sample mean $\bar{X} = \frac{1}{n}\sum X_i$, we use $M_{\bar{X}}(t) = M_{\sum X_i}(t/n)$:

$$M_{\bar{X}}(t) = \exp\left(n\mu \cdot \frac{t}{n} + \frac{1}{2}\, n\sigma^2 \cdot \frac{t^2}{n^2}\right) = \exp\left(\mu t + \frac{1}{2} \cdot \frac{\sigma^2}{n} \cdot t^2\right).$$

This is the mgf of $N\left(\mu, \frac{\sigma^2}{n}\right)$. Therefore $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$. □

Key takeaway: The sample mean $\bar{X}$ is centered at the true population mean μ. Its variance σ^2/n shrinks as

n grows, meaning larger samples produce more precise estimates of μ.

1.5 The Central Limit Theorem (CLT)

The previous result required the population to be normal. The Central Limit Theorem is a much more powerful result that applies to *any* distribution.

Central Limit Theorem

Let $X_1, X_2, \ldots, X_n$ be independent random variables from the *same* distribution with mean μ and variance σ^2. Then the limiting form of the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}},$$

as $n \to \infty$, is the standard normal distribution.

Rule of thumb: The approximation is generally considered adequate when $n \geq 30$.

Professor's remark: "The CLT is the most important theorem in this course. It is what allows us to do all of statistics — without the CLT, we couldn't build confidence intervals or do hypothesis tests unless we knew the exact population distribution. Remember: the CLT applies to the **sample mean**, not to the raw data. A large sample does NOT make the data itself normal."

Why this is powerful: Even if the individual X_i follow a skewed, uniform, exponential, or other non-normal distribution, once we average enough of them together, the distribution of the average becomes approximately normal. This is the theoretical foundation for much of classical statistics.

Equivalent formulations of the CLT:

1. If X_i are i.i.d. from a distribution with mean μ and variance σ^2, then:

$$\bar{X} \overset{\text{approx}}{\sim} N\left(\mu,\ \frac{\sigma^2}{n}\right) \qquad \text{for large } n.$$

2. For the sum: if $\bar{X} \overset{\text{approx}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$, then:

$$\sum_{i=1}^{n} X_i \overset{\text{approx}}{\sim} N(n\mu,\ n\sigma^2).$$

Remark: $n \geq 30$ is a good cutoff for the approximation. For populations that are already close to normal, the CLT kicks in for much smaller n. For highly skewed populations, you may need larger n.

1.6 Data Descriptions: Quantiles, Boxplots, and QQ-Plots

1.6.1 Quantiles

A quantile of a sample (say x_p) is the value below which a specific fraction p of the data values fall. Quantiles are often described as **percentiles**.

Quartiles are special cases of quantiles:

- 1st quartile Q_1 (25th percentile),
- 2nd quartile Q_2 (50th percentile, i.e., the median),
- 3rd quartile Q_3 (75th percentile).

The **five-number summary** of a dataset consists of: min, Q_1, Q_2, Q_3, max.

1.6.2 Boxplot

A boxplot is a graphical display based on the five-number summary. It shows:

- a box from Q_1 to Q_3 (the **interquartile range**, IQR $= Q_3 - Q_1$),
- a line at the median inside the box,
- whiskers extending to the smallest and largest observations within $1.5 \times$ IQR of the quartiles,
- individual points (possible outliers) beyond the whiskers.

Understanding the Interquartile Range (IQR)

The **Interquartile Range (IQR)** is one of the most important measures of spread:

$$\boxed{\text{IQR} = Q_3 - Q_1}$$

What it captures: The IQR represents the range of the *middle 50%* of the data. Unlike the full range (max − min), the IQR is **resistant to outliers** because it ignores the extreme values in the tails.

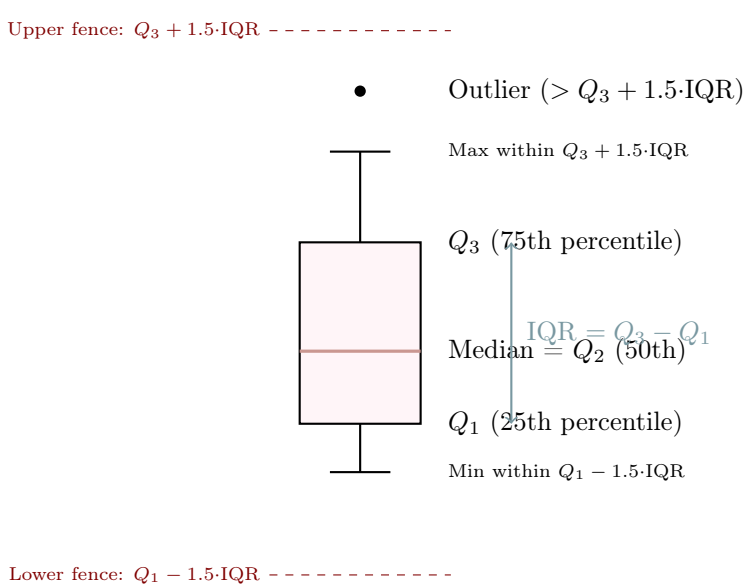
How to compute Q_1 and Q_3:

1. Sort the data in ascending order.
2. Find the median (Q_2): this splits the data into a lower half and an upper half.
3. Q_1 = median of the **lower half** of the data (below the overall median).
4. Q_3 = median of the **upper half** of the data (above the overall median).

Outlier detection using IQR:

- **Lower fence:** $Q_1 - 1.5 \times \text{IQR}$
- **Upper fence:** $Q_3 + 1.5 \times \text{IQR}$
- Any data point below the lower fence or above the upper fence is flagged as a **possible outlier**.
- The whiskers of the boxplot extend only to the most extreme data points *within* these fences, not to the fences themselves.

Professor's remark: "The IQR is preferred over the range or standard deviation when you suspect outliers in the data, because it only looks at the middle half. This makes the boxplot a robust summary."



1.6.3 Quantile Plot

A quantile plot displays data values on the vertical axis against the fraction of observations exceeded by the data value on the horizontal axis. Key properties:

- All data points are plotted (unlike a boxplot which summarizes).
- It is **descriptive**, not diagnostic.

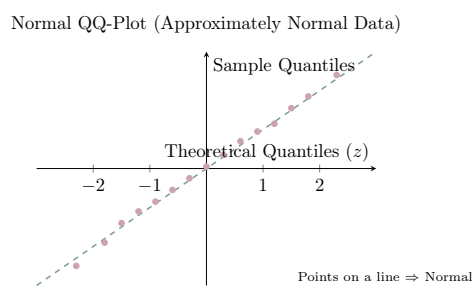
1.6.4 Normal Quantile-Quantile (QQ) Plot

The normal QQ-plot is a **diagnostic** tool used to determine whether a data set came from a normal distribution.

- Plot the data quantiles (vertical axis) against the corresponding quantiles of the standard normal distribution (horizontal axis).
- If the data are approximately normal, the points will fall approximately on a **straight line**.
- Systematic deviations from linearity indicate non-normality.
- **Outliers** appear as points that are far from the overall pattern.

Note on the textbook: The x-axis labels in the book's QQ-plot figure should be $-2, -1, 0, 1, 2$ (standard normal quantiles), not the values shown.

Professor's remark: "On the exam, you will be asked to interpret QQ-plots, not to construct them from scratch. The key patterns to know: if the points curve upward at the right end, the data are right-skewed; if they curve downward at the left end, the data are left-skewed; if they form an S-shape, the data have heavier tails than normal. A straight line means approximately normal."



1.6.5 Normal Probability Plot

Similar to the QQ-plot, but rather than corresponding z-scores we have corresponding standard normal percentiles and a special scale. The interpretation is the same:

- Look for a straight line — systematic deviations indicate non-normality.
- Outliers appear as points far from the overall pattern.

1.7 Chebyshev's Inequality

Theorem: Chebyshev's Inequality

The probability that X will assume a value within k standard deviations of the mean is at least $1-\frac{1}{k^2}$:

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}.$$

Interpretation: Chebyshev's inequality gives a lower bound on probability that works for **any** distribution (as long as the mean and variance exist). It does not require normality.

Proof (continuous case):

We start with the definition of variance:

$$\sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx.$$

Split the integral into three regions:

$$\sigma^2 = \int_{-\infty}^{\mu-k\sigma} (x-\mu)^2 f(x)\, dx + \int_{\mu-k\sigma}^{\mu+k\sigma} (x-\mu)^2 f(x)\, dx + \int_{\mu+k\sigma}^{\infty} (x-\mu)^2 f(x)\, dx.$$

Since the middle integral is non-negative, we can drop it to get a lower bound:

$$\sigma^2 \geq \int_{-\infty}^{\mu-k\sigma} (x-\mu)^2 f(x)\, dx + \int_{\mu+k\sigma}^{\infty} (x-\mu)^2 f(x)\, dx.$$

For $x \leq \mu - k\sigma$ or $x \geq \mu + k\sigma$, we have $|x-\mu| \geq k\sigma$, so $(x-\mu)^2 \geq k^2\sigma^2$. Substituting:

$$\sigma^2 \geq k^2\sigma^2 \left[\int_{-\infty}^{\mu-k\sigma} f(x)\, dx + \int_{\mu+k\sigma}^{\infty} f(x)\, dx \right].$$

Dividing both sides by $k^2\sigma^2$:

$$\int_{-\infty}^{\mu-k\sigma} f(x)\, dx + \int_{\mu+k\sigma}^{\infty} f(x)\, dx \leq \frac{1}{k^2}.$$

Therefore:

$$P(\mu - k\sigma < X < \mu + k\sigma) = 1 - \left[\int_{-\infty}^{\mu-k\sigma} f(x)\, dx + \int_{\mu+k\sigma}^{\infty} f(x)\, dx \right] \geq 1 - \frac{1}{k^2}. \qquad \square$$

1.8 R Code: QQ-Plots and CLT Demonstration

1.8.1 Creating Random Samples and QQ-Plots

```
# Creating random samples from a given distribution
```

```r
x <- rf(2000, 3, 27)       # random sample of size 2000 from F(3, 27)
x <- rnorm(100, 2, 100)    # random sample of size 100 from N(2, 100)

# Plotting a normal quantile-quantile plot
qqnorm(x)
qqline(x)

# For the normal probability plot (requires "e1071" package)
library(e1071)
probplot(x)    # draw a normal probability plot
```

1.8.2 Demonstrating the CLT

The idea is to:

1. Generate random data from a non-normal distribution (e.g., $F(3, 27)$, which is right-skewed).
2. Repeatedly draw samples of size n and compute the sample mean each time.
3. Plot the distribution of these sample means.
4. Observe that as n increases, the distribution of $\bar{X}$ becomes more and more normal.

```r
# Show that the F(3,27) distribution is right-skewed
x <- rf(2000, 3, 27)
hist(x)

# CLT demonstration
n <- 4    # try different values: n=4, 10, 30, 100
xbar <- array(0, dim = c(0, 1000))  # vector of 1000 zeros

for(i in 1:1000) {
  xbar[i] <- mean(rf(n, 3, 27))     # sample mean of n obs from F(3,27)
}

hist(xbar)       # should look more normal as n increases
qqnorm(xbar)     # should fall on a straight line
qqline(xbar)
```

Key observation: When $n = 4$, the histogram of $\bar{X}$ is still somewhat skewed. As n increases to 30 or more, the histogram looks approximately normal and the QQ-plot becomes nearly linear. This is the Central Limit Theorem in action.

1.9 True or False: Conceptual Questions

The following true/false questions test understanding of sampling distributions and the CLT:

1. **As n (sample size) increases, the mean of the sampling distribution of $\bar{X}$ decreases.**

   False. The mean of $\bar{X}$ is always μ, regardless of n. The sample size does not affect the center of the sampling distribution.

2. **As n increases, the standard deviation of the sampling distribution of $\bar{X}$ decreases.**

   True. The standard deviation of $\bar{X}$ is $\sigma/\sqrt{n}$, which decreases as n grows. Larger samples produce more precise estimates.

3. **The mean $\bar{X}$ of a random sample of size 4 from a left-skewed distribution is approximately normally distributed.**

   False. The CLT requires $n \geq 30$ (approximately) for the normal approximation to be adequate. With $n = 4$ from a skewed distribution, $\bar{X}$ will not be approximately normal.

4. **The distribution of the proportion of successes $\hat{p}$ in a sufficiently large sample is approximately normal with mean p and std. dev. $\sqrt{np(1-p)}$.**

   False. The standard deviation of $\hat{p} = X/n$ is $\sqrt{p(1-p)/n}$, not $\sqrt{np(1-p)}$. The quantity $\sqrt{np(1-p)}$ is the standard deviation of the *count* X, not the proportion $\hat{p}$.

5. **If $\bar{X}$ is the mean of a simple random sample of size 9 from $N(500, \sigma^2 = 18)$, then $\bar{X}$ has a normal distribution with mean 500 and variance 36.**

   False. Since $X_i \sim N(500, 18)$, we have $\bar{X} \sim N(500, 18/9) = N(500, 2)$. The variance of $\bar{X}$ is $18/9 = 2$, not 36.

6. **A large sample from a skewed population distribution will have an approximately normal histogram.**

   False. The *data itself* will still look skewed. It is the *sampling distribution of* $\bar{X}$ that becomes approximately normal, not the distribution of the raw data.

7. **The mean of a population will be normally distributed if the population is quite large.**

   False. The population mean μ is a fixed number (a parameter), not a random variable. It does not have a distribution. The CLT applies to the *sample mean* $\bar{X}$, not to the population mean.

8. **The average blood cholesterol level recorded in a simple random sample of 100 students from a large population will be approximately normally distributed.**

   True. By the CLT, with $n = 100 \geq 30$, the sample mean $\bar{X}$ is approximately normal regardless of the shape of the population distribution.

1.10 Examples: Applying the CLT to the Sum of Random Variables

1.10.1 Example 1: Airline Passenger Weights

Problem. Weights of airline passengers are known to have a distribution with mean $\mu = 75\,\text{kg}$ and standard deviation $\sigma = 10\,\text{kg}$. A certain plane has a passenger weight capacity of $7700\,\text{kg}$. What is the probability that a flight of 100 passengers will exceed the capacity?

Setup. Let X_i denote the weight (in kg) of the i-th passenger, so $X_i \sim (\mu = 75, \sigma = 10)$. Since passengers are randomly selected, $X_1, \ldots, X_{100}$ are i.i.d. We want:

$$P\left(\sum_{i=1}^{100} X_i > 7700\right).$$

Intuition. If each passenger weighs 75 kg on average, then 100 passengers should total around $100 \times 75 = 7500\,\text{kg}$. The question is how likely the total exceeds 7700 kg.

Solution. By the CLT (since $n = 100 \geq 30$):

$$\sum_{i=1}^{100} X_i \overset{\text{approx}}{\sim} N(100 \times 75,\ 100 \times 100) = N(7500,\ 10000)\,.$$

The variance of the sum is $n\sigma^2 = 100 \times 10^2 = 10{,}000$, so the standard deviation is $\sqrt{10{,}000} = 100$.

$$P\left(\sum_{i=1}^{100} X_i > 7700\right) = P\left(Z > \frac{7700 - 7500}{\sqrt{100 \times 100}}\right) = P\left(Z > \frac{200}{100}\right) = P(Z > 2) \approx \boxed{0.0228.}$$

Conclusion: There is approximately a 2.28% chance the flight exceeds the weight capacity.

1.11 Inferences on the Population Mean

So far, we have used probability to compute things given known parameters (e.g., "given $\mu = 75$, find $P(\ldots)$"). Now we flip the question: given observed data, what can we infer about the unknown μ?

1.11.1 The Setup

We have a sample $X_1, \ldots, X_n$ that are i.i.d. with unknown mean μ and known (or estimable) standard deviation σ. We observe the sample mean $\bar{X}$ and ask: is the data consistent with a particular assumed value of μ?

By the CLT (when n is large), we know:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

1.11.2 Example 2: Cylindrical Component Diameters

Problem. A manufacturing process produces cylindrical components. An engineer conjectures that the population mean diameter is $\mu = 5\,\text{mm}$. A random sample of $n = 100$ parts is selected, yielding $\bar{x} = 5.027\,\text{mm}$. It is known that $\sigma = 0.1\,\text{mm}$. Does the sample support the engineer's conjecture?

Note: In practice, σ is rarely "known" exactly. Here we take it as a given assumption — perhaps it was established from extensive prior production data. The case where σ must be estimated from the sample leads to t-procedures covered later.

Setup. Let X_i be the diameter of the i-th component (in mm). Then $X_1, \ldots, X_{100}$ are i.i.d. with unknown μ and $\sigma = 0.1$. By the CLT:

$$\bar{X} \sim N\left(\mu,\ \frac{0.1^2}{100}\right) = N\left(\mu,\ \frac{0.01}{100}\right).$$

The standard deviation of $\bar{X}$ is $\dfrac{0.1}{\sqrt{100}} = 0.01$.

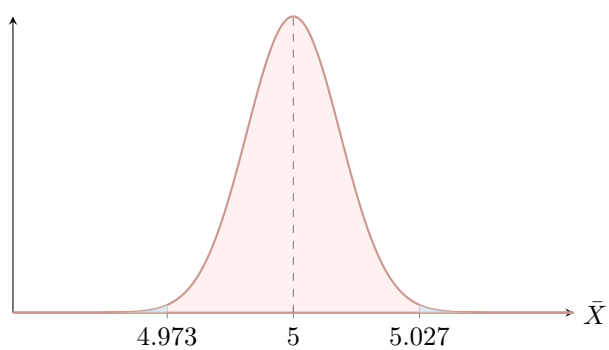
Assuming $\mu = 5$. How unlikely is it to observe a sample mean that deviates from 5 by 0.027 mm or more?

$$P\left(\bar{X} \leq 4.973 \ \text{ or } \ \bar{X} \geq 5.027 \,\middle|\, \mu = 5\right) = 1 - P(4.973 < \bar{X} < 5.027)$$

By symmetry:

$$= 2\,P\left(Z > \frac{5.027 - 5}{0.1/10}\right) = 2\,P\left(Z > \frac{0.027}{0.01}\right) = 2\,P(Z > 2.7) \approx \boxed{0.007.}$$

Visualization. The sampling distribution of $\bar{X}$ under $\mu = 5$ is shown below. The shaded tails represent the probability of observing a sample mean as extreme as or more extreme than $\bar{x} = 5.027$.



Interpretation. A probability of $0.007 = 0.7\%$ means that, *if the true mean really were 5 mm*, observing a sample mean as far as 5.027 mm from 5 would be a very rare event. Since 0.007 is well below the conventional threshold of $\alpha = 0.05$, we say the observed data **does NOT support** the engineer's conjecture that $\mu = 5\,\text{mm}$.

On the significance threshold $\alpha = 0.05$: In science and engineering, the conventional cutoff for declaring a result statistically significant is $\alpha = 0.05$ (5%). This is sometimes called the "risk bound" or significance level. If the probability of seeing data at least as extreme as what we observed—assuming the null hypothesis is true—falls below α, we reject the null hypothesis. Here, $0.007 < 0.05$, so we reject $\mu = 5$.

Professor's remark: "The p-value is NOT the probability that H_0 is true. It is the probability of seeing data at least as extreme as what we observed, **given that H_0 is true**. Small p-value means the data are surprising under H_0, so we reject H_0. Think of it like a rare event — if something very unlikely happened under the assumption, maybe the assumption is wrong."

1.12 Sampling Distribution of the Difference Between Two Averages

Theorem: Two-Sample Difference of Means

Let independent random samples of sizes n_1 and n_2 be drawn from two populations (discrete or continuous) with means μ_1, μ_2 and variances σ_1^2, σ_2^2, respectively. If n_1 and n_2 are sufficiently large, then:

$$\bar{X}_1 - \bar{X}_2 \overset{\text{approx}}{\sim} N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

Proof sketch. By the CLT applied to each sample independently:

$$\bar{X}_1 \overset{\text{approx}}{\sim} N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right) \quad \text{(from CLT)} \qquad \text{and} \qquad \bar{X}_2 \overset{\text{approx}}{\sim} N\left(\mu_2, \frac{\sigma_2^2}{n_2}\right) \quad \text{(from CLT)}.$$

Since the samples are independent, and from Chapter 7 we know that *linear combinations of independent normal random variables are normal*, we get:

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right). \qquad \square$$

Professor's remark: "When comparing two populations, always set up the problem under H_0. If H_0 says the means are equal, then $\mu_1 - \mu_2 = 0$, and your distribution is centered at 0. The variance is always $\sigma_1^2/n_1 + \sigma_2^2/n_2$ — the variances **add**, they never subtract, because variability always increases when you combine independent sources."

1.12.1 Example 3: Comparing Paint Drying Times

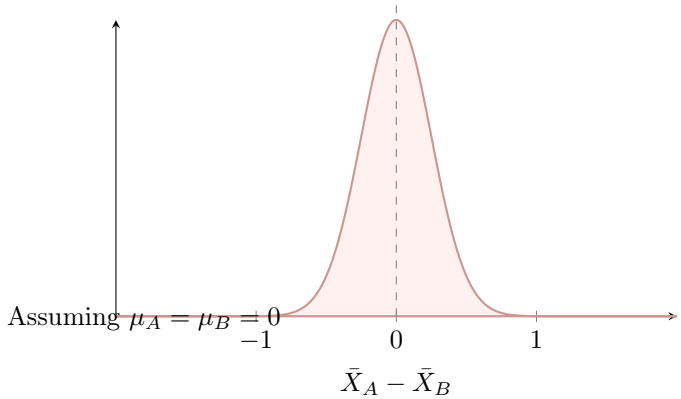
Problem. In a manufacturing study, 32 specimens are painted with Paint A and another 32 are painted with Paint B. The drying time in hours is recorded for each specimen. It is known that $\sigma = 1$ hour for both paints. The observed difference in sample averages was $\bar{x}_A - \bar{x}_B = 1$. Is there evidence that $\mu_A \neq \mu_B$?

Setup. Under the assumption that $\mu_A = \mu_B$ (i.e., the paints are equally fast), the sampling distribution of the difference is:

$$\bar{X}_A - \bar{X}_B \overset{\text{approx}}{\sim} N\left(\mu_A - \mu_B,\ \frac{\sigma_A^2}{32} + \frac{\sigma_B^2}{32}\right) = N\left(0,\ \frac{1}{32} + \frac{1}{32}\right) = N\left(0,\ \frac{1}{16}\right).$$

The standard deviation of $\bar{X}_A - \bar{X}_B$ is $\frac{1}{4}$.

Visualization. Under $\mu_A = \mu_B = 0$, the distribution is centered at 0 with the observed difference of ± 1 shown in the tails:



Calculation.

$$P\Big(\bar{X}_A - \bar{X}_B > 1 \;\Big|\; \mu_A = \mu_B\Big) + P\Big(\bar{X}_A - \bar{X}_B < -1 \;\Big|\; \mu_A = \mu_B\Big)$$
$$= 2\,P\left(\frac{(\bar{X}_A - \bar{X}_B) - 0}{1/4} > \frac{1-0}{1/4}\right) = 2\,P(Z > 4) < 0.0001.$$

Conclusion. The probability of observing a difference as large as 1 hour (in absolute value) under the null hypothesis is less than 0.0001. This is far below any reasonable significance level. We **reject** $\mu_A = \mu_B$ and conclude there is strong evidence of a difference in mean drying times.

1.13 Summary: Hypothesis Testing Procedure

Step-by-Step Hypothesis Testing (from the Board)

Step 1: State the hypotheses.

- H_0: Null hypothesis (the "status quo" or "nothing special" claim).
- H_a: Alternative hypothesis (what we suspect or want to show).

Step 2: Compute the test statistic.

- σ known, or large n: $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$
- σ unknown, small n (normal pop.): $T = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t(n-1)$
- Two-sample (equal var.): $T = \dfrac{\bar{X}_1 - \bar{X}_2}{S_p\sqrt{1/n_1 + 1/n_2}} \sim t(n_1 + n_2 - 2)$
- Variance test: $F = \dfrac{S_1^2}{S_2^2} \sim F(n_1 - 1, n_2 - 1)$

Step 3: Determine the rejection region OR compute the p-value.
Step 4: Make a decision. Reject H_0 if test stat falls in rejection region (or $p < \alpha$).
Step 5: State the conclusion in context.

Professor's remark: "On the exam, always show all five steps. Even if you get the arithmetic wrong, showing the correct procedure earns partial credit. Also remember: the pooled variance formula (s_p^2) is only used when you have evidence that the two population variances are equal — typically from an F-test."

Confidence Intervals vs. Hypothesis Tests

Connection: A two-sided $(1-\alpha)$ confidence interval and a two-sided α-level test are equivalent:

- If μ_0 falls **inside** the CI $\Rightarrow$ fail to reject H_0 at level α.
- If μ_0 falls **outside** the CI $\Rightarrow$ reject H_0 at level α.

This is why you can sometimes answer a hypothesis test question by looking at a confidence interval (as in Practice 8.3b).

1.14 Practice Problems

Practice 8.1 — CLT: Sum of Random Variables (adapted from 2014 exam)

Problem. Suppose that $X_1, \ldots, X_{200}$ are independent continuous random variables with common density function

$$f(x) = 2(1-x) \quad \text{for } 0 \le x \le 1, \quad 0 \text{ otherwise.}$$

Define $S = X_1 + \cdots + X_{200}$.

(a) Compute the mean and variance of X_i.

(b) Using an appropriate method, find an approximation for $P(S > 70)$.

(c) Now define $Y_i = 1$ if $X_i > 0.9$ and $Y_i = 0$ otherwise. Set $T = Y_1 + \cdots + Y_{200}$. Compute $P(T \ge 3)$.

Solution.

(a)

$$E(X) = \int_0^1 x \cdot 2(1-x)\,dx = 2\int_0^1 (x - x^2)\,dx = 2\left[\frac{1}{2} - \frac{1}{3}\right] = \frac{1}{3}.$$

$$E(X^2) = \int_0^1 x^2 \cdot 2(1-x)\,dx = 2\left[\frac{1}{3} - \frac{1}{4}\right] = \frac{1}{6}.$$

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{1}{6} - \frac{1}{9} = \frac{1}{18}.$$

(b) By the CLT, S is approximately normal with:

$$E(S) = 200 \times \frac{1}{3} = \frac{200}{3} \approx 66.67, \qquad \mathrm{Var}(S) = 200 \times \frac{1}{18} = \frac{200}{18} \approx 11.11.$$

$$P(S > 70) = P\left(Z > \frac{70 - 200/3}{\sqrt{200/18}}\right) = P\left(Z > \frac{70 - 66.67}{\sqrt{11.11}}\right) = P(Z > 1.00) \approx 0.1587.$$

(c) First, find $P(X_i > 0.9)$:

$$P(X_i > 0.9) = \int_{0.9}^1 2(1-x)\,dx = \left[-(1-x)^2\right]_{0.9}^1 = 0 - (-(0.1)^2) = 0.01.$$

So $T = \sum_{i=1}^{200} Y_i \sim \mathrm{Bin}(200,\ 0.01)$.
Exact computation:

$$\begin{aligned} P(T \ge 3) &= 1 - P(T=0) - P(T=1) - P(T=2) \\ &= 1 - \binom{200}{0}(0.01)^0(0.99)^{200} - \binom{200}{1}(0.01)^1(0.99)^{199} - \binom{200}{2}(0.01)^2(0.99)^{198} \\ &\approx 1 - 0.1340 - 0.2707 - 0.2720 = \boxed{0.3233.} \end{aligned}$$

Practice 8.2 — Hypothesis Test on Population Mean (adapted from 2015 exam)

Problem. A program was run 6 times with randomly chosen data sets. The sample mean and sample standard deviation of execution times were $\bar{x} = 228$ ms and $s = 14$ ms, respectively.

(a) Compute a 95% confidence interval for the true mean execution time. State your assumptions.

(b) It is claimed that the overall mean execution time is less than 230 ms. Do the data support this claim? Use $\alpha = 0.05$.

Solution.
(a) The sample is small ($n = 6$) and σ is unknown, so we assume the population is normally distributed and use the t-distribution with $n - 1 = 5$ degrees of freedom.
$t_{5;\,0.025} = 2.571$. The 95% CI is:

$$\bar{x} \pm t_{n-1;\,\alpha/2}\,\frac{s}{\sqrt{n}} = 228 \pm 2.571 \times \frac{14}{\sqrt{6}} = 228 \pm 14.7 = (213.3,\ 242.7)\,\text{ms}.$$

(b) Hypotheses: H_0: $\mu = 230$ vs. H_a: $\mu < 230$.

$$t_{\text{stat}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{228 - 230}{14/\sqrt{6}} = -0.35.$$

The critical value is $-t_{5;\,0.05} = -2.015$. Since $t_{\text{stat}} = -0.35 > -2.015$, we **fail to reject** H_0. The data do not provide sufficient evidence to support the claim.

Practice 8.3 — Two-Sample Inference on Means (adapted from 2015 exam)

Problem. A company supporting two types of computer networks measures downtime (hours/year):

	Type 1	Type 2
Sample size	10	9
Sample mean	83.5	60.4
Standard deviation	34.2	22.3

(a) Test whether the two populations have different variances at $\alpha = 0.10$.

(b) Assuming equal variances, is there a significant difference in mean downtime? Use $\alpha = 0.10$.

Solution.
(a) H_0: $\sigma_1^2 = \sigma_2^2$ vs. H_a: $\sigma_1^2 \neq \sigma_2^2$.

$$F_{\text{stat}} = \frac{s_1^2}{s_2^2} = \frac{34.2^2}{22.3^2} = \frac{1169.64}{497.29} \approx 2.352.$$

Critical value: $F_{(9,8);\,0.05} = 3.39$. Since $2.352 < 3.39$, we **fail to reject** H_0. There is no significant evidence of unequal variances.
(b) Using the pooled t-test (equal variance assumed):

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{9 \times 34.2^2 + 8 \times 22.3^2}{17} \approx 853.24.$$

$t_{17;\,0.05} = 1.740$. The 90% CI for $\mu_1 - \mu_2$:

$$(83.5 - 60.4) \pm 1.740\sqrt{853.24\left(\frac{1}{10} + \frac{1}{9}\right)} = 23.1 \pm 22.85 \approx (0.25,\ 45.95).$$

Since the CI does not contain 0, we **reject** H_0 at $\alpha = 0.10$ and conclude there is a significant difference in mean downtime.

Practice 8.4 — Inference and Power (adapted from 2015 exam)

Problem. Define $X_1, \ldots, X_{100}$ to be independent Normal r.v.'s with unknown mean μ and variance 25. Test H_0: $\mu = 10$ vs. H_1: $\mu \neq 10$ using test statistic $\bar{X} = \frac{1}{100}\sum_{i=1}^{100} X_i$. Reject H_0 if $\bar{X} \geq 10.98$ or $\bar{X} \leq 9.02$.

(a) Find the significance level of the test.

(b) If the true value is $\mu = 10.5$, find the power of the test.

Solution.
(a) Under H_0, $\bar{X} \sim N\big(10,\ \frac{25}{100}\big) = N(10, 0.25)$, so $\mathrm{SD}(\bar{X}) = 0.5$.

$$\alpha = P(\bar{X} \geq 10.98) + P(\bar{X} \leq 9.02) = 2\,P\left(Z > \frac{10.98 - 10}{0.5}\right) = 2\,P(Z > 1.96) = 2(0.025) = \boxed{0.05.}$$

(b) Under H_1: $\mu = 10.5$, $\bar{X} \sim N(10.5, 0.25)$.

$$\begin{aligned}
\text{Power} &= P(\bar{X} \geq 10.98 \mid \mu = 10.5) + P(\bar{X} \leq 9.02 \mid \mu = 10.5) \\
&= P\left(Z > \frac{10.98 - 10.5}{0.5}\right) + P\left(Z < \frac{9.02 - 10.5}{0.5}\right) \\
&= P(Z > 0.96) + P(Z < -2.96) \\
&\approx 0.1685 + 0.0015 = \boxed{0.17.}
\end{aligned}$$

Practice 8.5 — Sampling Distribution of Difference (two paints)

Problem. Suppose specimens are painted with Paint A or Paint B. We know $\sigma_A = \sigma_B = 1$ hour, and independent samples of size $n_A = n_B = 32$ are taken. The observed difference is $\bar{x}_A - \bar{x}_B = 1$.

(a) State the null and alternative hypotheses for testing whether the paints differ in mean drying time.

(b) Under H_0, what is the sampling distribution of $\bar{X}_A - \bar{X}_B$?

(c) Compute the p-value and state your conclusion.

Solution.
(a) H_0: $\mu_A = \mu_B$ (equivalently $\mu_A - \mu_B = 0$) vs. H_a: $\mu_A \neq \mu_B$.

(b) Under H_0:

$$\bar{X}_A - \bar{X}_B \sim N\left(0,\ \frac{1}{32} + \frac{1}{32}\right) = N\left(0,\ \frac{1}{16}\right), \quad \text{SD} = \frac{1}{4}.$$

(c)

$$p\text{-value} = 2\,P\big(\bar{X}_A - \bar{X}_B > 1\big) = 2\,P\left(Z > \frac{1-0}{1/4}\right) = 2\,P(Z > 4) < 0.0001.$$

Since $p < 0.05$, we **reject** H_0 and conclude that the paints have different mean drying times.

Practice 8.6 — Hypothesis Test: Binomial with Normal Approximation (adapted from 2014 exam)

Problem. Suppose that X is a Binomial random variable with $n = 100$ and unknown p. We want to test H_0: $p = 0.4$ against H_1: $p > 0.4$. We reject H_0 if $X \geq 48$. Use normal approximations throughout.

(a) Find the significance level (α) for this test.

(b) Suppose that the true value of p is 0.55. Find the power of the test.

Solution.
(a) Under H_0, $X \sim \text{Bin}(100, 0.4)$, so $\mu = 40$, $\sigma = \sqrt{100 \times 0.4 \times 0.6} = \sqrt{24}$.

$$\alpha = P(X \geq 48 \mid p = 0.4) \approx P\left(Z \geq \frac{47.5 - 40}{\sqrt{24}}\right) = P\left(Z \geq \frac{7.5}{4.899}\right) = P(Z \geq 1.531) \approx \boxed{0.063.}$$

(b) Under H_1, $X \sim \text{Bin}(100, 0.55)$, so $\mu = 55$, $\sigma = \sqrt{100 \times 0.55 \times 0.45} = \sqrt{24.75}$.

$$\text{Power} = P(X \geq 48 \mid p = 0.55) \approx P\left(Z \geq \frac{47.5 - 55}{\sqrt{24.75}}\right) = P\left(Z \geq \frac{-7.5}{4.975}\right) = P(Z \geq -1.508) \approx \boxed{0.934.}$$

Practice 8.7 — Assessing Normality (adapted from 2013 exam)

Problem. A machine produces metal rods. A random sample of 21 rods is selected and the diameter of each is measured. Assuming normality, a 95% confidence interval was constructed.

(a) How might you assess the assumption of normality? State two different methods. For each, indicate what would suggest normality and what would suggest non-normality.

Solution.

(a) Two methods to assess normality:

Method 1: Normal QQ-plot (or normal probability plot). Plot the data quantiles against the theoretical normal quantiles. If the points fall approximately on a straight line, this supports the normality assumption. Systematic curvature or deviations from linearity suggest non-normality.

Method 2: Formal hypothesis test (e.g., Shapiro–Wilk test, chi-squared goodness-of-fit

test). If you reject the null hypothesis (that the data come from a normal distribution), there is evidence of non-normality. If you fail to reject, the normality assumption is not contradicted.

Practice 8.8 — Sample Size for Power (adapted from 2013 exam)

Problem. Suppose $X_1, \ldots, X_n$ are i.i.d. $N(\mu, 1)$. We test H_0: $\mu = 10$ vs. H_a: $\mu \neq 10$ at the 5% level, rejecting if $\bar{X} > 10 + 1.96/\sqrt{n}$ or $\bar{X} < 10 - 1.96/\sqrt{n}$. If the true value of μ is 10.1, how large must n be for the test to have power at least 0.90?

Solution.

Under H_a: $\mu = 10.1$:

$$\text{Power} = P\left(\bar{X} > 10 + \frac{1.96}{\sqrt{n}} \;\middle|\; \mu = 10.1\right) + P\left(\bar{X} < 10 - \frac{1.96}{\sqrt{n}} \;\middle|\; \mu = 10.1\right).$$

The second term approaches 0 quickly. Focusing on the dominant term:

$$P\left(Z > \frac{10 + 1.96/\sqrt{n} - 10.1}{1/\sqrt{n}}\right) = P\big(Z > 1.96 - 0.1\sqrt{n}\big) \geq 0.90.$$

For $P(Z > c) = 0.90$, we need $c = -1.28$ (approximately). So:

$$1.96 - 0.1\sqrt{n} = -1.28 \implies 0.1\sqrt{n} = 3.24 \implies \sqrt{n} = 32.4 \implies n \approx 1050.$$

Therefore, n must be at least approximately $\boxed{1050}$ (the exact answer with full normal probabilities is $n = 1051$).

Chapter 2

Chapter 8 (cont.): Laws of Large Numbers and Sampling Distribution of S^2

2.1 Law of Large Numbers — Informal Introduction

Setup: Let $X_1, X_2, \ldots$ be i.i.d. random variables with mean μ and variance σ^2. Define the sample mean:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Key observation (from CLT):

$$E(\bar{X}_n) = \mu, \qquad \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

As $n \to \infty$, $\mathrm{Var}(\bar{X}_n) = \sigma^2/n \to 0$.

Since the variance of $\bar{X}_n$ shrinks to zero, the sample mean is becoming tightly concentrated around μ. This is the heart of the Law of Large Numbers.

So: $\bar{X}_n \to E(\bar{X}_n) = \mu$.

2.2 Weak Law of Large Numbers (WLLN)

Theorem: Weak Law of Large Numbers

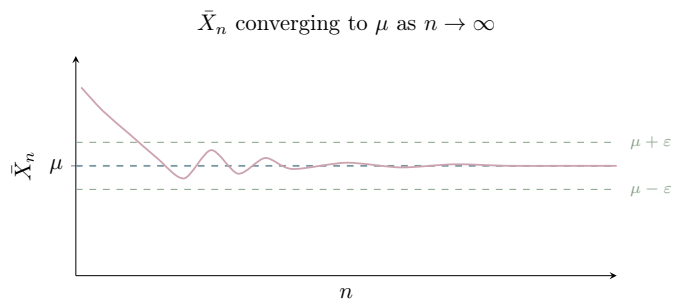
Let $X_1, X_2, \ldots$ be i.i.d. random variables with $E(X_i) = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Then for every $\varepsilon > 0$:

$$\lim_{n\to\infty} P\big(|\bar{X}_n - \mu| < \varepsilon\big) = 1.$$

That is, $\bar{X}_n$ **converges to μ in probability**.

What this means: No matter how small a tolerance ε you pick, as long as you take a large enough sample, the probability that $\bar{X}_n$ falls within ε of the true mean μ can be made as close to 1 as you like.

Proof: The WLLN is proved using Chebyshev's inequality (see Section 2.5).



2.3 Strong Law of Large Numbers (SLLN)

Theorem: Strong Law of Large Numbers

Under the same conditions as the WLLN:

$$P\Big(\lim_{n\to\infty} |\bar{X}_n - \mu| < \varepsilon\Big) = 1.$$

That is, $\bar{X}_n$ converges to μ **almost surely** (with probability 1).

Difference from the WLLN:

- **WLLN** (convergence in probability): For any fixed $\varepsilon > 0$, the probability that $\bar{X}_n$ is far from μ goes to 0. But different samples could still occasionally stray for a while.
- **SLLN** (almost sure convergence): The event that $\bar{X}_n \to \mu$ happens for sure — the limit of the sequence equals μ with probability 1. This is a stronger statement: it says the path $\bar{X}_n$ actually settles down to μ ("happens for sure").

Professor's remark: "For the exam, you mainly need to know the WLLN and how to prove it using Chebyshev. The SLLN is conceptual — know the difference, but you won't be asked to prove it. The key takeaway: more data = better estimate. The sample mean will converge to the population mean."

2.4 Markov's Inequality

Theorem: Markov's Inequality

If X is a **non-negative** random variable with $E(X) < \infty$ and $a > 0$, then:

$$P(X \geq a) \leq \frac{E(X)}{a}.$$

Intuitive interpretation: No more than $\frac{1}{5}$ of the population can have more than 5 times the average income. (Set $a = 5\mu$: $P(X \geq 5\mu) \leq \mu/(5\mu) = 1/5$.)

Proof (continuous case):

$$\begin{aligned} E(X) &= \int_0^\infty x f(x)\,dx = \int_0^a x f(x)\,dx + \int_a^\infty x f(x)\,dx \\ &\geq \int_a^\infty x f(x)\,dx \geq \int_a^\infty a\, f(x)\,dx = a\, P(X \geq a). \qquad \square \end{aligned}$$

Note: Markov's inequality is the foundational inequality — Chebyshev's inequality is derived from it.

Professor's remark: "Markov requires only $X \geq 0$ and $E(X) < \infty$ — very minimal assumptions. Chebyshev also needs the variance to exist, so it uses more information and gives tighter bounds. On the exam, always check: does the problem give you only a mean (use Markov) or both mean and variance (use Chebyshev)?"

2.5 Chebyshev's Inequality

Theorem: Chebyshev's Inequality

The probability that X assumes a value within k standard deviations of the mean is at least $1 - \frac{1}{k^2}$:

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}.$$

Example: For $k = 2$: $P(\mu - 2\sigma < X < \mu + 2\sigma) \geq 1 - \frac{1}{4} = \frac{3}{4}$.

Key point: Unlike the 68–95–99.7 rule (which requires normality), Chebyshev's inequality applies to **any distribution** with finite mean and variance. It is a weaker bound but universally applicable.

Proof (continuous case): See the uploaded proof notes. The key steps are:

1. Write $\sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx$ and split into three regions.
2. Drop the middle integral (it is non-negative), keeping only the outer tails.
3. Use $(x-\mu)^2 \geq k^2\sigma^2$ on the tails to replace and simplify.
4. Divide by $k^2\sigma^2$ and complement to get the inequality.

Proof of the WLLN via Chebyshev's inequality:

Apply Chebyshev's inequality to $\bar{X}_n$, which has mean μ and variance σ^2/n:

$$P\big(|\bar{X}_n - \mu| \geq \varepsilon\big) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \xrightarrow{n\to\infty} 0.$$

Therefore $P(|\bar{X}_n - \mu| < \varepsilon) \to 1$. $\square$

2.6 Sampling Distribution of S^2

Recall the sample variance:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

2.6.1 Theorem: Chi-Squared Distribution of S^2

Theorem: Sampling Distribution of S^2

If S^2 is the sample variance of a random sample of size n taken from a **normal** distribution with variance σ^2, then:

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{\sigma^2} \sim \chi^2(n-1),$$

the chi-squared distribution with $(n-1)$ **degrees of freedom**.

Why $n-1$ degrees of freedom? We lose one degree of freedom because we estimated the mean μ with $\bar{X}$. The deviations $X_i - \bar{X}$ satisfy $\sum(X_i - \bar{X}) = 0$, so only $n-1$ are free to vary independently.

Professor's remark: "This result is extremely important — it is the basis for testing claims about population variance. The key thing to remember: this result **requires normality**. Unlike the CLT which works for any distribution, the chi-squared distribution of S^2 specifically needs the underlying population to be normal. Also remember that $\bar{X}$ and S^2 are independent when sampling from a normal distribution — this is a unique property of the normal."

2.6.2 Proof of the Chi-Squared Result

Step 1 — Decomposition identity: We start by writing:

$$\sum_{i=1}^{n}(X_i - \mu)^2 = \sum_{i=1}^{n}\left[(X_i - \bar{X}) + (\bar{X} - \mu)\right]^2.$$

Expanding:

$$= \sum_{i=1}^{n}(X_i - \bar{X})^2 + \sum_{i=1}^{n}(\bar{X} - \mu)^2 + 2(\bar{X} - \mu)\underbrace{\sum_{i=1}^{n}(X_i - \bar{X})}_{=0} = \sum_{i=1}^{n}(X_i - \bar{X})^2 + n(\bar{X} - \mu)^2.$$

Step 2 — Divide by σ^2: Substituting $(n-1)S^2$ for $\sum(X_i - \bar{X})^2$:

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2 = \frac{(n-1)S^2}{\sigma^2} + \frac{(\bar{X}-\mu)^2}{\sigma^2/n}.$$

Step 3 — Identify the pieces: Since $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$:

- $\dfrac{X_i - \mu}{\sigma} \sim N(0,1)$, so $\left(\dfrac{X_i-\mu}{\sigma}\right)^2 \sim \chi^2(1)$.
- $\dfrac{1}{\sigma^2}\displaystyle\sum_{i=1}^{n}(X_i-\mu)^2 =$ sum of n independent $\chi^2(1)$ r.v.'s $\sim \chi^2(n)$.
- $\dfrac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$, so $\dfrac{(\bar{X}-\mu)^2}{\sigma^2/n} \sim \chi^2(1)$.

By the additivity of chi-squared (if $A \sim \chi^2(n)$ and $B \sim \chi^2(1)$ are independent and $A = C + B$, then $C \sim \chi^2(n-1)$):

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1). \qquad \square$$

*Additionally, this shows that $\bar{X}$ and S^2 are **independent** of each other when sampling from a normal population — a remarkable fact unique to the normal distribution.*

2.6.3 Chi-Squared Additivity Theorem

Theorem: Additivity of Chi-Squared

If $X_1 \sim \chi^2(\nu_1),\ X_2 \sim \chi^2(\nu_2), \ldots, X_k \sim \chi^2(\nu_k)$ are independent, then:

$$\sum_{i=1}^{k} X_i \sim \chi^2(\nu_1 + \nu_2 + \cdots + \nu_k).$$

This can be easily proved using the chi-squared moment generating function.

2.7 Summary: Sampling Distributions from a Normal Population

Key Results Summary (from the Board)

1. If $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ for $i = 1, \ldots, n$, then $\bar{X} \sim N\left(\mu, \dfrac{\sigma^2}{n}\right)$ **exactly**.

2. **CLT**: If $X_i \overset{\text{iid}}{\sim}$ any distribution with mean μ and variance σ^2, then $\bar{X} \overset{\text{approx}}{\sim} N\left(\mu, \dfrac{\sigma^2}{n}\right)$ for $n \geq 30$.

3. If $X_{1i} \sim N(\mu_1, \sigma_1^2)$ (sample of n_1) and $X_{2i} \sim N(\mu_2, \sigma_2^2)$ (sample of n_2) independently, then:

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

2.8 Worked Example: Chi-Squared Test for Variance

Example — Car Battery Lifetimes

Problem. A manufacturer guarantees its car batteries will last, on average, 3 years with $\sigma \leq 1$ year. Five batteries are tested and found to have lifetimes (in years): 1.9, 2.4, 3.0, 3.5, 4.2. Is the claim $\sigma^2 \leq 1$ correct?

Given: $n = 5$, so $n - 1 = 4$ degrees of freedom. Assume $\sigma^2 = 1$ (null hypothesis).

Step 1 — Compute S^2:

$$\bar{x} = \frac{1.9 + 2.4 + 3.0 + 3.5 + 4.2}{5} = \frac{15.0}{5} = 3.0.$$

$$S^2 = \frac{(1.9-3)^2 + (2.4-3)^2 + (3.0-3)^2 + (3.5-3)^2 + (4.2-3)^2}{4} = \frac{1.21 + 0.36 + 0 + 0.25 + 1.44}{4} = \frac{3.26}{4} = 0.815.$$

Step 2 — Compute the test statistic:

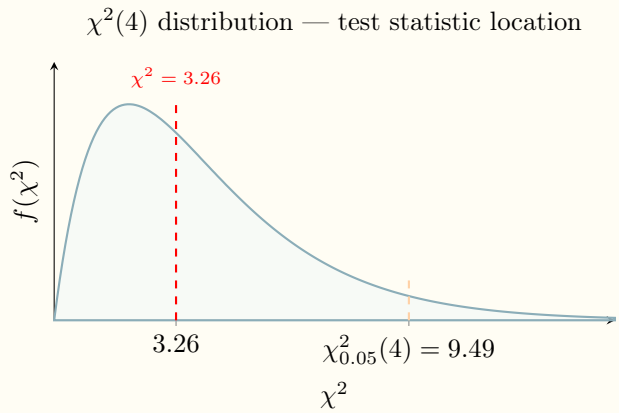
Assume $\sigma^2 = 1$ and lifetime is normally distributed.

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{4 \times 0.815}{1} = 3.26 \ \sim\ \chi^2(4).$$

Step 3 — Find the p-value:

We want $P(\chi^2(4) > 3.26)$. From the $\chi^2(4)$ table, $P(\chi^2 > 0.5153) \approx$ high. Since 3.26 is well within the central mass of a $\chi^2(4)$ distribution, we **fail to reject** H_0.

Conclusion: The data do not provide sufficient evidence to reject the manufacturer's claim that $\sigma^2 \leq 1$.



2.9 Practice Problems

Practice — LLN and Chebyshev

Problem 1. Let $X_1, X_2, \ldots$ be i.i.d. with mean $\mu = 5$ and variance $\sigma^2 = 4$.

(a) Using Chebyshev's inequality, find a lower bound on $P(3 < \bar{X}_{100} < 7)$.

(b) What does the WLLN tell you about $\bar{X}_n$ as $n \to \infty$?

(c) If in addition the $X_i \sim N(5, 4)$, compute the exact probability $P(3 < \bar{X}_{100} < 7)$.

Solution.
(a) $\bar{X}_{100}$ has mean 5 and $\mathrm{Var}(\bar{X}_{100}) = 4/100 = 0.04$, so $\mathrm{SD}(\bar{X}_{100}) = 0.2$.
The interval $(3, 7)$ is $\mu \pm 2$, i.e. $\mu \pm k\sigma_{\bar{X}}$ where $k = 2/0.2 = 10$.
By Chebyshev: $P(3 < \bar{X}_{100} < 7) \geq 1 - 1/10^2 = 1 - 0.01 = \mathbf{0.99}$.
(b) The WLLN says $\bar{X}_n \to 5$ in probability; for any $\varepsilon > 0$, $P(|\bar{X}_n - 5| < \varepsilon) \to 1$.
(c) $\bar{X}_{100} \sim N(5, 0.04)$ exactly (since X_i normal). Standardize: $z = (3-5)/0.2 = -10$ and $z = (7-5)/0.2 = 10$. $P(3 < \bar{X}_{100} < 7) = P(-10 < Z < 10) \approx \mathbf{1.0000}$.

Practice — Markov and Chebyshev Comparison

Problem 2. $X \geq 0$ has mean 4.

(a) Use Markov's inequality to bound $P(X \geq 20)$.

(b) If additionally $\mathrm{Var}(X) = 2$, use Chebyshev's inequality to bound $P(|X - 4| \geq 4)$.

(c) Which inequality gives a tighter bound in (b)? Why?

Solution.
(a) $P(X \geq 20) \leq E(X)/20 = 4/20 = \mathbf{0.2}$.
(b) Chebyshev: $P(|X - 4| \geq 4) \leq \mathrm{Var}(X)/4^2 = 2/16 = \mathbf{0.125}$.

(c) Chebyshev gives a tighter bound (0.125 vs. Markov applied to $|X-4|^2$ would give the same; Chebyshev uses variance information that Markov does not).

Practice — Sampling Distribution of S^2 (adapted from 2012 exam)

Problem 3. A random sample of size $n = 10$ is taken from a normal population with variance $\sigma^2 = 2$. Find $P(S^2 > 3.5)$.

Solution.
$\dfrac{(n-1)S^2}{\sigma^2} = \dfrac{9S^2}{2} \sim \chi^2(9)$.
We want:

$$P(S^2 > 3.5) = P\left(\frac{9S^2}{2} > \frac{9 \times 3.5}{2}\right) = P(\chi^2(9) > 15.75).$$

From the $\chi^2(9)$ table: $\chi^2_{0.05}(9) = 16.92$ and $\chi^2_{0.10}(9) = 14.68$.
Since $14.68 < 15.75 < 16.92$, we have $0.05 < P(\chi^2(9) > 15.75) < 0.10$.
$\therefore P(S^2 > 3.5) \approx \mathbf{0.07}$ (between 5% and 10%).

Practice — Chi-Squared Additivity (adapted from past exam)

Problem 4. Suppose $X_1 \sim \chi^2(3)$ and $X_2 \sim \chi^2(5)$ are independent.

(a) What is the distribution of $X_1 + X_2$?

(b) Find $P(X_1 + X_2 > 15.507)$.

(c) Find $E(X_1 + X_2)$ and $\mathrm{Var}(X_1 + X_2)$.

Solution.
(a) By additivity: $X_1 + X_2 \sim \chi^2(3+5) = \boldsymbol{\chi^2(8)}$.
(b) $P(\chi^2(8) > 15.507) = 0.05$ (this is the standard 5% critical value for 8 d.f.).
(c) $E(\chi^2(\nu)) = \nu$ and $\mathrm{Var}(\chi^2(\nu)) = 2\nu$:

$$E(X_1 + X_2) = 3 + 5 = \mathbf{8}, \qquad \mathrm{Var}(X_1 + X_2) = 2(3) + 2(5) = \mathbf{16}.$$